Round 1 Report: Stage 1 (to 15 Oct)

RoboKnights SemiCon · AXI4-Stream → PCIe 6.0 bridge · Aditya (interconnect), Swaray (link layer), Kshitij (protocol-to-PHY) · Oct 24, 2026 · @Aditya

Summary

By 15 Oct each of the three layers had working RTL with passing unit testbenches, and a first top level (`Final/system_top.v`) chained all three with a PHY loopback. A beat could travel from the AXI4-Stream ports through arbitration, the clock crossing, session gating, Flit framing and the full Gray / PAM4 lane chain, and come back as a reconstructed Flit with clean CRC / FEC status.

That chain was a **connectivity** result, not yet a data-correctness one: no TLP packetizer sat between the interconnect and the link layer, several link-layer blocks (tag manager, credit manager, packetizers, address translation, RX engine) were tested alone but not wired in, and the link–PHY interface had no back-pressure. Those gaps defined Round 2.

Layer	Owner	State at 15 Oct
Interconnect	Aditya	Complete; 4 real bugs found and fixed; unit + integration benches pass (128 / 128 beats, 0 errors)
Link layer	Swaray	Data-link core wired (write buffer, replay buffer, TX/RX Flit framing, DLLP parser); transaction-layer blocks unit-tested but not integrated
Protocol-to-PHY	Kshitij	Complete lane chain bit-exact in loopback; reset / config / MSI-X control plane; formal proofs for Gray and PAM4; cocotb 4 / 4
System top	All	Three layers chained; smoke tests pass

Sources: the Stage 1 design report (`docs/report_round1.pdf`), git history 6–13 Sep, and the team's working chats as merged in the project ledger (`claude/SEMICON_LEDGER.md`, §2). Swaray's own list of Stage 1 bug fixes is still to come and will be added to the link-layer section.

Problem and scope decisions

The bridge moves data from FPGA-fabric AXI4-Stream producers into host memory over a PCIe 6.0 link for two workloads: **SmartNIC** packet ingestion and **ML inference** output streaming. Both are strictly device-to-host writes.

Two decisions made early shaped everything after:

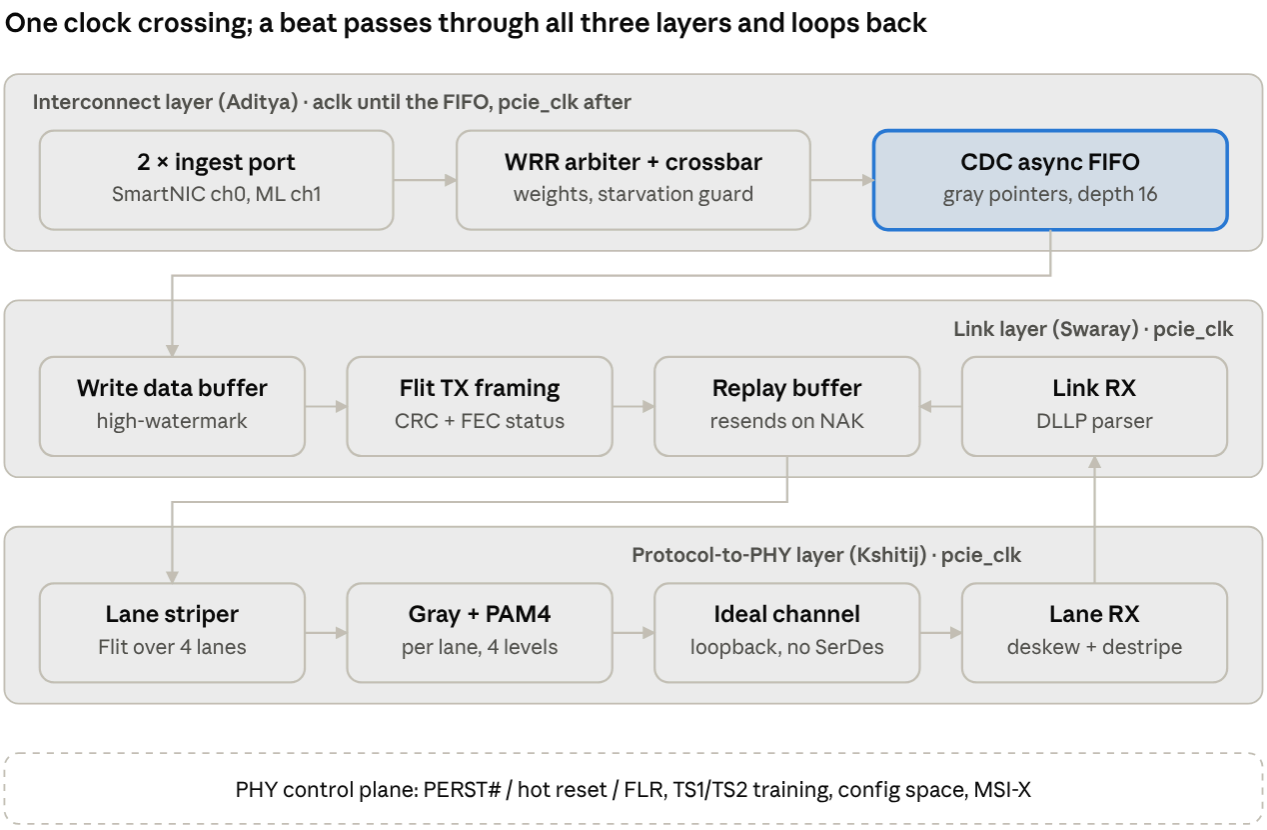
1. **AXI4-Stream only (AD-SCOPE-1).** After reading the problem slides, the interconnect scope was cut to two independent AXI4-Stream ingest ports. Dropped: AXI4 memory-mapped AW/W/B and AR/R ports, the read-completion clock crossing and a per-beat 64-bit address in TUSER.
2. **Address split across layers (AD-SCOPE-2).** The interconnect never handles a 64-bit address. It produces only a per-channel running byte offset; the link layer forms `real_addr = base_addr_ch[N] + session_offset_ch[N]`, where the base addresses are host-written registers in the PHY layer's config space (BAR0 0x20 / 0x28). One MSI-X vector and one host buffer per channel. This keeps the interconnect datapath 24 bits wide instead of 64 and stores each address once.

Questions still open at the time of that design note: the session-arm handshake, the bound on a session's size, the overflow policy, Flit- vs beat-aligned offsets and shared back-pressure. Each was settled during Stage 1 or Round 2 (see the last section).

Cross-layer tickets exchanged during Stage 1:

Ref	From → to	Question	Outcome
AD-ST-A / B	Swaray → Aditya	TUSER latched per burst; no interleaving of the two channels	Closed with no RTL change: the link layer keeps one Flit buffer per stream
AD-KT-1	Kshitij → Aditya	`session_valid` a pulse or a level?	Level in Stage 1; became a clean clock-crossed pulse in Round 2 (REV-I3)
AD-KT-2	Kshitij → Aditya	Compile-time `SLICE_SIZE`	Held (would break the port list); became runtime per-channel limit registers in Step 4 (S4-7)
AD-KT-3–6	Kshitij → Aditya	Overflow policy, TUSER fields, Flit alignment, shared back-pressure	Answered; matched the RTL

Architecture and interfaces



Stage 1 architecture · 3 layers, 1 clock crossing, PHY loopback

The only crossing between the AXI clock (`aclk` , target ≤ 250 MHz) and the PCIe clock (`pcie_clk`) is the gray-code FIFO in the interconnect layer; everything after it runs on `pcie_clk` . With no real SerDes in the repository, the reconstructed Flit is looped straight back into the link layer's receive side, which exercises the whole encode / decode chain in simulation.

Interface	Signals	Notes
Interconnect → link	`m_tdata` 512, `m_tkeep` 64, `m_tuser` 26 = `{tlast_flag, session_offset[23:0], stream_id}` , `m_tlast` , `m_tvalid` / `m_tready`	FWFT handshake. Arbitration is per beat, so the two channels interleave and the link layer keeps independent per-channel state.
Session arm (host → interconnect, per channel)	`base_addr_in` 64, `tc_in` 3, `session_arm` (write-1-to-set), `session_valid` (level)	3-state FSM IDLE → LATCH → ACTIVE latches the address into shadow registers.

Interface	Signals	Notes
Link ↔ PHY	`m_phy_tx_valid/data/sof/eof` (1 / 256 / 1 / 1), same on RX, `crc_err`, `fec_err`	One 256-bit word = one Flit. **No ready / back-pressure** in Stage 1.
Control-plane sideband	reset requests, TS1/TS2 bytes, CfgRd/CfgWr, MSI-X table access, DMA-complete pulse → MSI-X MWr	Plain top-level ports on `system_top.v`.

Interconnect layer (Aditya)

Modules

Module	Function
`axi4s_ingest_port` (× 2)	AXI4-Stream slave per channel (0 = SmartNIC, 1 = ML). Byte-accurate session offset from a popcount of TKEEP, look-ahead overflow guard with a sticky `session_overflow` cleared only by the next session start, TLAST ends the session. Separate `req` (to the arbiter) and `accepted` (handshake really completed).
`rr_arbiter_2ch`	Weighted round-robin with runtime weights (default 1:1), per-channel credit refill, starvation guard after `STARVE_CYCLES` (64). Grants are registered, so no combinational path reaches the 512-bit data mux.
`cdc_fifo_async`	Gray-code asynchronous FIFO, the only clock crossing. Depth 16, 2-flop synchronisers, first-word-fall-through read, `almost_full` at 75% used as shared back-pressure for both channels. Word = 512 data + 64 keep + 1 id + 24 offset + 1 last = 602 bits.
`interconnect_top`	Two ingest ports → crossbar mux → arbiter select → FIFO → output. TUSER repacked as one bus to match the link layer's contract.
`session_arm_ctrl`, Stage 1 `bridge_top`	Per-channel session-arm FSM; first cross-team point wiring the interconnect into the link layer's write data buffer.

Bugs found and fixed in Stage 1

Ref	Severity	Bug	Fix
R1-IC-1	FUNC	A beat held under back-pressure was written into the FIFO again every cycle, because the FIFO write was gated on the raw grant.	The ingest port exports `accepted`; FIFO write = `(grant0 && ch0_accepted) | (grant1 && ch1_accepted)`.
R1-IC-2	BLOCKER	The arbiter's joint round-done (both credits zero) deadlocked when only one channel requested.	Independent per-channel round-done that treats an idle channel as exhausted.
R1-IC-3	FUNC	Registered FIFO read gave stale data on the first reads; AXI4-Stream needs data valid with TVALID.	First-word-fall-through read.
R1-IC-4	BLOCKER	The full flag depended on a next pointer gated by the full flag itself: a combinational loop that hung the simulator.	Unconditional look-ahead pointer compared with the synchronised read pointer.

Testbench problems found along the way

- **Missing beat that wasn't (AD-TB-SAMPLE-1).** The integration bench reported one beat per channel missing (59 / 60, then 63 / 64). The RTL was correct: the driver sampled `s_tready` after the clock edge had already updated it, counted a beat as accepted one edge early and advanced its data. Proved by parsing the VCD and counting FIFO pointer transitions (writes = reads), then fixed by checking the handshake with zero delay. An intermediate wrong "fix" (settle delays on the debug counters) briefly misreported loss inside the FIFO; it was dropped in favour of VCD ground truth.
- **Last write dropped (AD-TB-RACE-1).** `tb_cdc_fifo_async` cleared `wr_en` on the sampling edge; fixed with a settle delay.
- **Rule written down for every later bench (AD-VERIF-SAMPLE-RULE):** observe settled DUT state after a delay, but check handshakes (`tready` , `grant`) with zero delay at the edge. The link layer's registered `s_tready` can hit the same trap.

Tests: unit benches for the ingest port (offset tracking, TLAST, overflow guard, back-pressure), arbiter (50/50, 3:1, starvation guard, single requester, mutual exclusion) and FIFO (7 ns / 13 ns unrelated clocks, in-order, no loss), and a fully parameterised integration

bench with pointer-gated logs. Final run: **128 / 128 beats,** **`accepted0 = accepted1 = 64`**, **`wr_cnt = rd_cnt = 128`**, **0 errors.** A write-to-read latency formula for the FIFO, `lag ≈ N · max(0, Tr − Tw) + k · Tr` (k ≈ 2–3), was derived and checked against waveforms.

Constraints: async clock groups, 2-cycle multicycle on the gray synchronisers, placeholder periods (`.xdc` , `.sdc` , OpenLane config). OpenLane was not run in Stage 1 (no PDK set up).

Link layer (Swaray)

Wired into the Stage 1 chain

Module	Function
`write_data_buffer`	Merge buffer right after the interconnect's FIFO; back-pressure from a configurable high-watermark. End point of the Stage 1 `bridge_top`.
`replay_buffer`	Stores sent Flits and replays them on NAK. Passed TUSER through combinationally rather than storing it.
`link_layer_tx` / `link_layer_rx`	Per-channel Flit assembly with CRC / FEC framing and per-channel ready, because the arbiter interleaves both channels on one bus.
DLLP parser	Parses ACK / NAK from the receive stream and feeds sequence numbers back to the replay buffer.
`data_link_layer_top`	Wraps TX / RX, replay buffer and DLLP parser; 256-bit PHY-facing bus plus `crc_err` / `fec_err`.

Built and unit-tested, not yet in the chain: tag manager, credit manager, TLP packetizers (`tlp_packetizer` , `tlp32_to_axis_packetizer`), address translation, AR request splitter, AXI–PCIe reorder buffer, TX engine, PCIe RX engine, FEC encoder / decoder, Flit framer.

Tests (all passing): tag allocation / recycling / timeout, credit accounting, TLP header generation against known-good bytes (two packetizer benches), out-of-order completion reassembly, 4 KB boundary splitting, BAR decode with UR on miss, replay-buffer NAK retransmit, and loopback integration (`tb_link_layer_loopback` , `tb_top_pipeline`).

Bug fixes in Stage 1: commits of 6–13 Sep record module bug fixes and edge cases. The itemised list is waiting on Swaray's ledger return and will be added here.

What the 19 Oct review later found in this layer (replay buffer with no ACK source, sequence number overwriting data bits, 40 bits of every fourth beat dropped in Flit packing, FEC with no correction) is Round 2 work and is covered in the Round 2 report.

Protocol-to-PHY layer (Kshitij)

Module	Function
`gray_enc` / `gray_dec`	2-bit Gray encode / decode, exact inverses (formally proved).
`pam4_map` / `pam4_demap`	Gray symbol to one of four levels {−3, −1, +1, +3} and back.
`byte_to_symbol` / `symbol_to_byte`	Byte ↔ four 2-bit PAM4 symbols.
`flit_lane_striper` / `flit_lane_destriper`	256-bit Flit across 4 lanes, byte-interleaved over 8 cycles, and back.
`tx_elastic_deskew`	Per-lane elastic FIFOs with a shared priming threshold (striper emits a byte per lane per cycle; the lane pipe needs 4 cycles per byte).
`flit_sof_strip`	Flit-mode SOF token (0xFB) detection and 8-cycle payload pass-through. Known limit: 0xFB inside genuine payload.
TS ordered-set detector, lane-reversal detector / latch	TS1 / TS2 detection per lane, majority vote on lane reversal, locked on first training.
`pcie_reset_controller`	PERST#, hot reset, FLR. `datapath_rst_n` is deliberately not gated on link-up (training lives in that reset domain); `axi_rst_n` releases only after training and a first UpdateFC.
`pcie_config_space` , `msix_controller`	Type 0 header, up to 6 BARs, Device Control / Link Status; MSI-X table up to 2048 vectors with per-vector masking and PBA.
`pcie_phy_link_datapath`	Synthesizable wrapper of the lane chain with a 4-state auto-SOF FSM (IDLE → SOF → PUSH → DRAIN) replacing what had only been testbench stimulus.

Tests: `tb_end_to_end_datapath` (striper → Gray / PAM4 → ideal channel → demap → lane swap → SOF strip → destriper) bit-exact against the original Flit; SymbiYosys proofs `gray.sby`, `pam4.sby`, `verify_reset.sby`; cocotb `test_pcie_link_top.py` 4 / 4 (reset keeps config / MSI-X quiet, link-up makes config live, BAR sizing and writes, MSI-X masked / unmasked).

Issues known at the end of Stage 1: the 0xFB-in-payload limit of `flit_sof_strip`; modules duplicated across several files with a port-name mismatch, and a `sext_rst_n` typo in a system bench (both cleaned up in Round 2 as REV-P5 and REV-TB-1). Kshitij's working chat for this period was not part of the returns, so any further Stage 1 fixes are in `protocol_PHY_layer/CHANGELOG.md` (Updates 1–8).

System integration and verification results

`Final/system_top.v` instantiates, in order: the Stage 1 `bridge_top` (interconnect + write data buffer) → `data_link_layer_top` → `pcie_phy_link_datapath`, with the PHY output looped back into the link layer's receive side, plus `pcie_link_top` for the control plane. `datapath_rst_n` drives both the link and PHY datapaths, and the locked lane-reversal decision drives the datapath's lane swap instead of a tied-off placeholder.

The auto-SOF FSM captures each Flit into a small input FIFO (the link–PHY interface has no ready, so overflow is flagged on `tx_fifo_overflow` rather than lost silently), sends one SOF cycle, pulses the striper, and waits for the destriper before taking the next Flit: one Flit in flight at a time.

Bench	What it checks	Result
Interconnect unit benches (3)	Ingest, arbiter, FIFO as above	PASS
`tb_interconnect_top`	Two channels at once across independent clocks; stream id and offset sequencing	PASS, 128 / 128, 0 errors
`tb_bridge_smoke`	Two beats into channel 0; data and packed TUSER bit-for-bit at the write buffer output	PASS
`tb_system_top_smoke`	One beat in; a reconstructed Flit out of the PHY loopback with SOF / EOF, no CRC / FEC / FIFO-overflow flag (2,000-cycle timeout against ~100+ cycles per Flit)	PASS (connectivity)

Bench	What it checks	Result
`tb_end_to_end_datapath`	PHY lane chain bit-exact	PASS
PHY formal (`gray` , `pam4` , `verify_reset`)	Inverses, round trip, reset sequencing	Proved
PHY cocotb	Control plane	4 / 4
Link-layer unit benches (9)	As listed in the link-layer section	PASS individually

`tb_system_top_smoke` is a connectivity test, not a payload check: with no TLP packetizer between the interconnect and the link layer yet, the reconstructed Flit does not carry a real TLP header.

Open at the end of Stage 1, and how each closed

Every gap listed at the end of Stage 1 was closed in Round 2 except the deliberate scope limit on the receive direction. The references point into the project ledger and the Round 2 report.

Open item at 15 Oct	Closed by	How
No TLP packetizer between interconnect and link layer; transaction-layer blocks not integrated	Ticket rounds T1 / T2 (20 Oct), Step 1 G1–G6	Full `bridge_top` → `link_layer_top` → `pcie_phy_top` chain with real TLPs, checked end to end (144 / 144 TLPs)
No ready / back-pressure on the link → PHY interface	REV-Seam-C / REV-P3, then S4-3	PHY Flit FIFO + `phy_tx_ready` ; later a wide PHY (8 B / lane / clock)
One Flit in flight; link drains 1 beat / 20 clocks	S4-1, S4-2, S4-8	512-bit TLP engine, merging to max payload, 16-DW / clock RX; ML throughput 0.06 → 6.17 GB/s
Session-arm clock crossing (`session_valid` level from the PHY clock)	REV-I3	`pulse_sync` start pulse into `aclk`
Overflow abort sent no session end downstream (AD-IC-OVF-NOEND)	F1-FIX	Zero-length TLAST marker; also fixed the 24-bit wrap in the overflow compare

Open item at 15 Oct	Closed by	How
Compile-time session size question (AD-KT-2)	S4-7	Runtime per-channel session-limit registers (BAR0 0x0C / 0x34)
Arbiter weights give bursts, not interleave	S4-7	Interleaved weighted round-robin, one beat per grant
FEC detected but did not correct; replay had no ACK source	REV-L3, REV-L8	ACK / NAK in DLP and control Flits; 3-way interleaved SEC-DED + CRC-64
Duplicated PHY modules, `sext_rst_n` bench typo	REV-P5, REV-TB-1, T1-K2..K5	One module per file; legacy copies moved out of the build
`flit_sof_strip` 0xFB-in-payload limit	K6 (KS-K6-IDLE / -RX)	Partly: no change to SOF matching itself, but logical idle plus between-Flit framing checks now detect a slipped boundary and retrain the link
Stage 2: synthesis, physical design, GDSII	Steps 2–4, GDS phase	FPGA 255 MHz with timing met; interconnect GDS DRC / LVS clean on SKY130; full-bridge GDS runs in progress
Receive direction ("Master & Slave")	REV-I5	Inbound MWr to BAR0 only, by design: both use cases are device-to-host